

INTERNSHIP PROJECT REPORT

Project 4

Industry-Oriented Generative AI Resume Screening and Skill Matching System

Project Title	Generative AI Resume Screening & Skill Matching System
Domain	Artificial Intelligence / HR Technology
Type	Industry-Oriented Internship Project
Tools Used	Python, OpenAI API / LLM APIs, Prompt Engineering, LangChain (optional), Jupyter Notebook
Target Roles	Prompt Engineer, AI Engineer, HR-Tech Developer

1. Problem Statement

Recruiters receive thousands of resumes for job openings and struggle with manual screening, skill matching, and shortlisting. Traditional keyword-based systems frequently miss relevant candidates by failing to understand context, synonyms, or transferable skills. The industry is increasingly adopting generative AI with advanced prompt engineering techniques to analyze resumes contextually and make smarter hiring decisions.

2. Objective

To design and build a generative AI system that intelligently analyzes resumes and matches candidates to job descriptions using well-crafted prompt design. The system aims to automate the screening pipeline — extracting skills, scoring relevance, and producing transparent shortlisting decisions — reducing recruiter workload while improving match quality.

3. Functional Requirements

The system must support the following core capabilities:

- Resume text extraction from uploaded documents
- Job description parsing and analysis
- Skill and experience extraction using engineered prompts
- Candidate–job relevance scoring
- Shortlisting logic based on AI-generated insights
- Bias-aware prompt design to ensure fair evaluation
- Result explanation and transparency for recruiter review
- End-to-end system workflow documentation

4. Tools & Technologies

Tool / Technology	Purpose
Python	Core programming language for pipeline development
OpenAI API / LLM APIs	Generative AI inference for resume and JD analysis
Prompt Engineering Techniques	Designing structured prompts for skill extraction & scoring
LangChain (optional)	Chaining prompt flows and managing LLM interactions
Jupyter Notebook	Interactive development, testing, and documentation

5. System Workflow

The project follows a structured pipeline with the following stages:

- Stage 1 — Input Collection: Upload resume (PDF/text) and job description.
- Stage 2 — Text Extraction: Parse and clean resume and JD content using Python.
- Stage 3 — Prompt Engineering: Design prompts to extract skills, experience, qualifications.
- Stage 4 — LLM Analysis: Send prompts to OpenAI/LLM API and retrieve structured responses.
- Stage 5 — Relevance Scoring: Compute candidate–JD match score based on AI output.
- Stage 6 — Shortlisting: Apply threshold logic and rank candidates.
- Stage 7 — Explanation & Report: Generate transparent reasoning for each decision.

6. Expected Outcomes

Upon successful completion of this project, the following outcomes are anticipated:

- Hands-on experience with AI-powered HR tools and real-world recruitment workflows.
- Deep understanding of contextual text analysis using Large Language Models (LLMs).
- A fully functional, industry-ready generative AI portfolio project.
- Job relevance demonstrated for roles including Prompt Engineer, AI Engineer, and HR-Tech Developer.
- Documented codebase and workflow ready for GitHub submission and future verification.

7. Submission & Storage Guidelines

All completed project files and supporting documents must be stored safely for future verification and submission. It is strongly advised to maintain backups using the following platforms:

- Google Drive — for document storage and easy sharing.
- GitHub (Recommended for Technical Domains) — for code versioning, collaboration, and portfolio visibility.

Ensure the repository is well-organized with a README file, Jupyter notebooks, prompt templates, and sample outputs clearly labeled.

Declaration

I hereby declare that this project has been carried out by me as part of my internship program. The work presented is original, and all references to tools, APIs, and frameworks have been duly acknowledged

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